ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET first paper. Section A (20): multiple choice, true or false and choosing terms from a list · Section B (80): robot drives, control and IoT, 3D printing a robot part, networks and fieldbus, articulated and SCARA robots with gripper torque and friction, microcontroller inputs and outputs and the TCP, and testing a prototype. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Use only a black or blue pen. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 Industry 4.0 is mainly about …

A steam-powered machines. B smart, connected machines that share data.
C assembly lines without electricity. D hand tools.

1.1.2 The space that a robot's end effector can reach is called its …

A payload. B degrees of freedom.
C repeatability. D work envelope.

1.1.3 In 3D printing, infill refers to …

A the inside structure printed within a part. B the outer wall of a part.
C the print bed. D the size of the nozzle.

1.1.4 Which network topology connects every device to one central switch?

A Bus B Ring C Star D Mesh

1.1.5 Which lightweight protocol is widely used to send IoT sensor data?

A MQTT B FTP C SMTP D POP3

1.1.6 A robot that moves along straight X, Y and Z rails is a …

A SCARA robot. B gantry (Cartesian) robot.
C articulated robot. D delta robot.

1.1.7 The largest mass that a robot can safely handle is its …

A reach. B accuracy. C speed. D payload.

1.1.8 Which sensor measures the rate of rotation (angular velocity)?

A LDR B Thermocouple
C Gyroscope D Load cell

1.1.9 STL is a file format used for …

A 3D-printing models. B spreadsheets.
C music. D PLC programs.

1.1.10 Which device sends data only to the port of the device it is meant for?

A Hub B Switch C Repeater D Splitter

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 Inverse kinematics calculates the joint angles needed to place the end effector at a target position.

1.2.2 Profibus and Modbus are examples of fieldbus protocols.

1.2.3 Supports are never needed when 3D-printing overhanging parts.

1.2.4 Wireless communication is always more reliable than wired communication in a factory.

1.2.5 Cybersecurity is important for robots that are connected to a network.

(5)

1.3 Choose ONE term from the list below for each of the following descriptions, and write it next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

articulated robot; latency; slicer; actuator; end effector; teach pendant; Ethernet; mesh network; LiDAR; payload; cloud computing; encoder

1.3.1 A hand-held device used to move a robot and record its positions

1.3.2 Software that cuts a 3D model into layers and produces the printer instructions

1.3.3 The delay between sending data and it being received on a network

1.3.4 A sensor that measures distance with laser light to map its surroundings

1.3.5 A network in which each node can pass on data for other nodes

(5)

[20]

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1 Robots can be driven by electric, hydraulic or pneumatic actuators. Give ONE advantage and ONE disadvantage of EACH type of drive. (6)

2.2 Explain the difference between point-to-point control and continuous-path control of a robot. (2)

2.3 Give ONE example of how the Internet of Things (IoT) is used in EACH of the following: (3)

2.3.1 A smart home

2.3.2 Smart farming

2.3.3 Smart health care

2.4 Explain ONE security risk of IoT devices and how it can be reduced. (2)

[13]

QUESTION 3

> A robotics workshop must design and 3D-print new fingers for a gripper that picks up glass jars. The fingers must be strong, must fit onto the existing gripper with screws, and must not scratch the glass.

3.1 List, in the correct order, the steps from designing the finger on a computer to having a finished printed part. (4)

3.2 Explain how the infill percentage and the layer height affect the strength, the printing time and the surface finish of the finger. (4)

3.3 Explain why tolerances must be allowed for when designing parts that must fit together. (2)

3.4 Name THREE types of 3D-printing filament and give ONE property of each. (3)

[13]

QUESTION 4

4.1 Explain the difference between a star topology and a bus topology, and give ONE advantage of each. (6)

4.2 Explain the following terms:

4.2.1 Bandwidth (2)

4.2.2 IP address (2)

4.3 Name TWO fieldbus protocols and TWO wireless technologies used in industrial automation. (4)

4.4 Differentiate between a router and a switch. (2)

[16]

QUESTION 5

5.1 Articulated robots are used widely in car factories.

5.1.1 How many axes does a typical articulated industrial robot have? (1)

5.1.2 Explain the terms reach and repeatability. (4)

5.2 A robot gripper holds a 12 kg part 0,4 m from the wrist joint. Calculate the torque that the wrist must provide to hold the part horizontally. (5)

5.3 A two-finger friction gripper must lift a 5 kg box. The coefficient of friction between the fingers and the box is 0,5. Calculate the minimum force each finger must apply. (4)

5.4 Give THREE applications of SCARA robots. (3)

[17]

QUESTION 6

6.1 Explain how the right-hand rule is used to show the X, Y and Z axes of a robot's coordinate system. (2)

6.2 Explain how a microcontroller reads an analogue sensor and controls the speed of a DC motor. (4)

6.3 Explain what is meant by debouncing a push-button input. (2)

6.4 Explain why a relay module is needed when a microcontroller switches a 230 V lamp. (2)

6.5 Explain what the tool centre point (TCP) is and why it must be set accurately. (3)

[13]

QUESTION 7: PRACTICAL ROBOTICS

7.1 Name TWO different views in which a circuit design can be presented in software such as Fritzing. (2)

7.2 Explain why a flowchart is drawn before a robot program is written. (2)

7.3 Name FOUR criteria that can be used to test a robot prototype. (4)

[8]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Speed: $v = d \div t$	Acceleration: $a = \Delta v \div \Delta t$
Force: $F = m \times a$	Weight: $W = m \times g$ ($g = 9{,}8\ m/s^2$)
Torque: $\tau = F \times r$	Friction: $F_f = \mu \times N$
Friction gripper: $F = W \div (\mu \times n)$, n = number of fingers	Work: $W = F \times s$; Power: $P = W \div t$
Ohm's law: $V = I \times R$	Electrical power: $P = V \times I$